

• Notes for lecture 3

§ 3.1 Some further applications to differentiation theory

Definition 1. A family $(k_t)_{t>0}$ of functions in $L^1(\mathbb{R}^d)$ is said to be an **approximate identity** if

(a) $\int_{\mathbb{R}^d} k_t(x) dx = 1$ for all $t > 0$,

(b) there exists an $M > 0$ such that

$$\int_{\mathbb{R}^d} |k_t(x)| dx \leq M \text{ for all } t > 0, \text{ and}$$

(c) for every $\delta > 0$ one has

$$\lim_{t \rightarrow 0^+} \int_{|x| > \delta} |k_t(x)| dx = 0.$$

Exercise. Let $k \in L^1(\mathbb{R}^d)$ be such that $\int_{\mathbb{R}^d} k(x) dx = 1$. Prove that the family $(k_t)_{t>0}$ of functions given by $k_t(x) := t^{-d} k(t^{-1}x)$, $x \in \mathbb{R}^d$, $t > 0$, is an approximate identity.

Theorem 2. Let $(k_t)_{t>0}$ be an approximate identity.

• If $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$ then

$$\lim_{t \rightarrow 0^+} \|k_t * f - f\|_{L^p(\mathbb{R}^d)} = 0$$

• If $f \in C_c(\mathbb{R}^d)$ then $\lim_{t \rightarrow 0^+} k_t * f = f$ uniformly.

Proof. Consider the case where $f \in L^p(\mathbb{R}^d)$, $1 \leq p < \infty$. By using property (a) of definition 1, we may write

$$k_t * f(x) - f(x) = \int_{\mathbb{R}^d} k_t(y) (f(x-y) - f(x)) dy.$$

Given an $\varepsilon > 0$, we may find a $\delta > 0$ such that if $|y| \leq \delta$ then

$$\| \tau_y f - f \|_{L^p(\mathbb{R}^d)} < \varepsilon, \text{ where } \tau_y f(x) := f(x-y), x \in \mathbb{R}^d.$$

Having fixed such a $\delta > 0$, choose $t_0 > 0$ sufficiently small such that

$$\int_{|y| > \delta} |k_t(y)| dy < \varepsilon \text{ for all } t \in (0, t_0).$$

Fix such a t_0 and take a $t \in (0, t_0)$. By using Minkowski's integral inequality (see below), one has

$$\begin{aligned} \|k_t * f - f\|_{L^p(\mathbb{R}^d)} &\leq \left(\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |k_t(y)| |f(x-y) - f(x)| dy \right)^p dx \right)^{1/p} \\ &\leq \int_{\mathbb{R}^d} |k_t(y)| \underbrace{\left(\int_{\mathbb{R}^d} |f(x-y) - f(x)|^p dx \right)^{1/p}}_{\| \tau_y f - f \|_{L^p(\mathbb{R}^d)}} dy \end{aligned}$$

and so,

$$\|k_t * f - f\|_{L^p(\mathbb{R}^d)} \leq A + B,$$

where

$$A := \int_{|y| \leq \delta} |k_t(y)| \| \tau_y f - f \|_{L^p(\mathbb{R}^d)} dy$$

$$\text{and } B := \int_{|y| > \delta} |k_t(y)| \| \tau_y f - f \|_{L^p(\mathbb{R}^d)} dy$$

To bound A , we use the definition of δ and property (b) in definition 1,

$$A \leq \varepsilon \int_{|y| \leq \delta} |k_t(y)| dy \leq \varepsilon \int_{\mathbb{R}^d} |k_t(y)| dy \leq M\varepsilon.$$

To bound B , observe that since $\|\tau_y f - f\|_{L^p(\mathbb{R}^d)} \leq 2\|f\|_{L^p(\mathbb{R}^d)}$ one has

$$B \leq 2\|f\|_{L^p(\mathbb{R}^d)} \int_{|y|>\delta} |k_t(y)| dy < 2\|f\|_{L^p(\mathbb{R}^d)} \cdot \varepsilon.$$

We have thus show that, for $f \in L^p(\mathbb{R}^d)$, for any given $\varepsilon > 0$ there exists a $t_0 > 0$, depending on f, ε (and $(k_t)_{t>0}$), such that $t \in (0, t_0)$ implies that

$$\|k_t * f - f\|_{L^p(\mathbb{R}^d)} \leq (M + 2\|f\|_{L^p(\mathbb{R}^d)}) \varepsilon.$$

• The proof of the uniform convergence when $f \in C_c(\mathbb{R}^d)$ is left as an exercise. ■

Theorem 3 (Minkowski's integral inequality). Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be two σ -finite measure spaces. If $p \in [1, \infty)$, then for every $A \times B$ -measure function f one has

$$\left(\int_Y \left| \int_X f(x, y) d\mu(x) \right|^p d\nu(y) \right)^{1/p} \leq \int_X \left(\int_Y |f(x, y)|^p d\nu(y) \right)^{1/p} d\mu(x).$$

Proof. Exercise. ■

Examples. 1) Let $P(x) := \frac{C_d}{(1 + |x|^2)^{d+1/2}}$, $x \in \mathbb{R}^d$,

where $C_d > 0$ is such that $\int_{\mathbb{R}^d} P(x) dx = 1$.

One can show that $C_d = \Gamma[(d+1)/2] / \pi^{(d+1)/2}$

-The family $(P_t)_{t>0}$ is an approximate identity, called the Poisson kernel.

- 2) If $k = V_d^{-1} \cdot \chi_{B(0,1)}$ with V_d being the volume of $B(0,1)$, then $(k_t)_{t>0}$ is an approximate identity. Note that

$$\begin{aligned} M_c(f)(x) &= \sup_{t>0} \frac{1}{|B(x,t)|} \int_{B(x,t)} |f(y)| dy \\ &= \sup_{t>0} \frac{1}{V_d t^d} \int_{B(0,t)} |f(x-z)| dz \\ &= \sup_{t>0} \frac{1}{V_d t^d} \int_{\mathbb{R}^d} |f(x-z)| \chi_{B(0,1)}\left(\frac{z}{t}\right) dz \\ &= \sup_{t>0} (|f| * k_t)(x). \end{aligned}$$

Theorem 4. Let g be a non-negative function on $[0, \infty)$ such that g has finitely many discontinuities and g is decreasing.

Consider the function k given by

$$k(x) := g(|x|), \quad x \in \mathbb{R}^d.$$

If $k \in L^1(\mathbb{R}^d)$, then every $f \in L^1_{loc}(\mathbb{R}^d)$ one has

$$\sup_{t>0} (|f| * k_t)(x) \leq \|k\|_{L^1(\mathbb{R}^d)} M_c(f)(x)$$

for a.e. $x \in \mathbb{R}^d$, where $k_t(y) := t^{-d} k(t^{-1}y)$, $y \in \mathbb{R}^d$, $t > 0$.

Proof. See pp. 91-92 in [2]. ■

Corollary 5. Let g be as in the statement of the previous theorem

Let $k \in L^1(\mathbb{R}^d)$ be such that $\int_{\mathbb{R}^d} k(x) dx = 1$ and $|k(x)| \leq g(|x|)$ for all $x \in \mathbb{R}^d$.

For $t > 0$, define $k_t(x) := t^{-d} k(t^{-1}x)$, $x \in \mathbb{R}^d$.

If $f \in L^p(\mathbb{R}^d)$, $p \in [1, \infty)$, then

$$\lim_{t \rightarrow 0^+} (k_t * f)(x) = f(x) \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Proof. Exercise. ■

Then d is a metric on $\mathcal{S}(\mathbb{R}^d)$ and the space $(\mathcal{S}(\mathbb{R}^d), d)$ is complete.

Definition 7. The space of bounded linear functionals on $\mathcal{S}(\mathbb{R}^d)$ is called the space of tempered distributions and is denoted by $\mathcal{S}'(\mathbb{R}^d)$.

By definition, a linear map u from $\mathcal{S}(\mathbb{R}^d)$ to $\mathbb{C}$ belongs to $\mathcal{S}'(\mathbb{R}^d)$ if, and only if,

$$\lim_{k \rightarrow \infty} \langle u, \varphi_k \rangle = 0 \quad \text{whenever} \quad \lim_{k \rightarrow \infty} d(\varphi_k, 0) = 0.$$

A very useful result is the following.

Theorem 8. A linear map u from $\mathcal{S}(\mathbb{R}^d)$ to $\mathbb{C}$ belongs to $\mathcal{S}'(\mathbb{R}^d)$ if, and only if, there exists a $C > 0$ and $k, l \in \mathbb{N}_0$ such that

$$|\langle u, \varphi \rangle| \leq C \sum_{\substack{|a| \leq k \\ |b| \leq l}} p_{a,b}(\varphi)$$

for all $\varphi \in \mathcal{S}(\mathbb{R}^d)$.

Proof. See p. 121 in [2]. ■

Examples. 1) The Dirac mass δ_0 at the origin is the linear map given by

$$\langle \delta_0, \varphi \rangle = \varphi(0), \quad \varphi \in \mathcal{S}(\mathbb{R}^d).$$

Note that $|\langle \delta_0, \varphi \rangle| \leq p_{0,0}(\varphi)$ and so, $\delta_0 \in \mathcal{S}'(\mathbb{R}^d)$

Remark. If $f \in L^p(\mathbb{R}^d)$, $p \in [1, \infty)$, then the function $u(x, t) := (f * P_t)(x)$, $(x, t) \in \mathbb{R}^d \times (0, \infty)$ is harmonic in the upper half plane $\mathbb{R}_+^{d+1} := \{(x, t) \in \mathbb{R}^d \times (0, \infty)\}$ (with $(P_t)_{t>0}$ being the Poisson kernel) and

$$\lim_{t \rightarrow 0^+} u(x, t) = f(x) \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Hence, $u(x, t)$ solves the Dirichlet problem

$$\begin{cases} \partial_t^2 u + \sum_{j=1}^d \partial_j^2 u = 0 & \text{on } \mathbb{R}_+^{d+1} \\ u(x, 0) = f(x) & \text{for a.e. } x \in \mathbb{R}^d \end{cases}$$

§ 3.2 The Schwartz class and tempered distributions

Definition 6. The Schwartz class on $\mathbb{R}^d$ is the set of all complex-valued functions $f \in C^\infty(\mathbb{R}^d)$ satisfying

$$p_{\alpha, \beta}(f) := \sup_{x \in \mathbb{R}^d} |x^\alpha \partial^\beta f(x)| < \infty$$

for all multi-indices α, β . The Schwartz class is denoted by $\mathcal{S}(\mathbb{R}^d)$.

Example. The function $f(x) = e^{-|x|^2}$, $x \in \mathbb{R}^d$, belongs to $\mathcal{S}(\mathbb{R}^d)$.

Let $\{p_i\}_{i \in \mathbb{N}}$ denote an enumeration of $\{p_{\alpha, \beta} : \alpha, \beta \text{ multi-indices in } \mathbb{N}_0^d\}$. Define $d: \mathcal{S}(\mathbb{R}^d) \times \mathcal{S}(\mathbb{R}^d) \rightarrow [0, \infty)$

$$d(f, g) := \sum_{j=1}^{\infty} 2^{-j} \cdot \frac{p_j(f-g)}{1 + p_j(f-g)}, \quad f, g \in \mathcal{S}(\mathbb{R}^d).$$

2) If $f \in L^p(\mathbb{R}^d)$, $1 \leq p \leq \infty$, define $u_f : \mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle u_f, \varphi \rangle := \int_{\mathbb{R}^d} f(x) \varphi(x) dx,$$

$\varphi \in \mathcal{S}(\mathbb{R}^d)$.

Exercise. Show that $u_f \in \mathcal{S}'(\mathbb{R}^d)$.

3) For $\varphi \in \mathcal{S}(\mathbb{R})$, define p.v. $(\frac{1}{x}) : \mathcal{S}(\mathbb{R}) \rightarrow \mathbb{C}$ by

$$\langle \text{p.v.}(\frac{1}{x}), \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon < |x| \leq 1} \frac{\varphi(x)}{x} dx + \int_{|x| > 1} \frac{\varphi(x)}{x} dx,$$

$\varphi \in \mathcal{S}(\mathbb{R})$.

Note that $\langle \text{p.v.}(\frac{1}{x}), \varphi \rangle$ is well-defined. Indeed, for the second term, we have

$$\left| \int_{|x| > 1} \frac{\varphi(x)}{x} dx \right| \leq \int_{|x| > 1} \frac{|x \varphi(x)|}{x^2} dx$$

$$\leq \left(\sup_{x \in \mathbb{R}} |x \varphi(x)| \right) \cdot \int_{|x| > 1} x^{-2} dx = 2 \rho_{1,0}(\varphi) < \infty.$$

For the first term, fix an $\varepsilon \in (0,1)$ and note that $1/x$ is odd on $[-1, -\varepsilon) \cup (\varepsilon, 1]$. Hence,

$$\int_{\varepsilon < |x| \leq 1} \frac{\varphi(x)}{x} dx = \int_{\varepsilon < |x| \leq 1} \frac{\varphi(x) - \varphi(0)}{x} dx$$

Observe that for all $\varepsilon \in (0,1)$, one has

$$\int_{\varepsilon < |x| \leq 1} \frac{|\varphi(x) - \varphi(0)|}{|x|} dx \leq \int_{\varepsilon < |x| \leq 1} \frac{|x| \|\varphi'\|_{L^\infty(\mathbb{R})}}{|x|} dx$$

$$\leq 2 \|\varphi'\|_{L^\infty(\mathbb{R})} = 2 \rho_{0,1}(\varphi) < \infty.$$

Hence, the limit exists and

$$|\text{p.v.}(\frac{1}{x})| \leq 2(\rho_{0,1}(\varphi) + \rho_{1,0}(\varphi)).$$

and so, p.v. $(\frac{1}{x}) \in \mathcal{S}'(\mathbb{R})$.

§ 3.2.1 Operators acting on $\mathcal{S}'(\mathbb{R}^d)$.

We shall extend some operators $A: \mathcal{S}(\mathbb{R}^d) \rightarrow \mathcal{S}(\mathbb{R}^d)$ to $\tilde{A}: \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathcal{S}'(\mathbb{R}^d)$.

1. Differentiation. If $\varphi, \psi \in \mathcal{S}(\mathbb{R}^d)$ then one has

$$\int_{\mathbb{R}^d} \partial^\alpha \varphi(x) \psi(x) dx = (-1)^{|\alpha|} \int_{\mathbb{R}^d} \varphi(x) \partial^\alpha \psi(x) dx$$

for all multi-indices α . Motivated by this identity, for $u \in \mathcal{S}'(\mathbb{R}^d)$ and a multi-index α , define $\partial^\alpha u: \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle \partial^\alpha u, \varphi \rangle := (-1)^{|\alpha|} \langle u, \partial^\alpha \varphi \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^d).$$

Exercise. Show that $\partial^\alpha u \in \mathcal{S}'(\mathbb{R}^d)$.

2. Convolution with a Schwartz function.

For $\eta \in \mathcal{S}(\mathbb{R}^d)$, we use the notation

$$\tilde{\eta}(x) := \eta(-x), \quad x \in \mathbb{R}^d.$$

If $\eta, \varphi, \psi \in \mathcal{S}(\mathbb{R}^d)$ then $\langle \eta * \psi, \varphi \rangle = \langle \psi, \varphi * \tilde{\eta} \rangle$.

Motivated by the identity above, if $u \in \mathcal{S}'(\mathbb{R}^d)$ and $\eta \in \mathcal{S}(\mathbb{R}^d)$, one defines $u * \eta: \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle u * \eta, \varphi \rangle := \langle u, \varphi * \tilde{\eta} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^d).$$

Theorem 9. If $u \in \mathcal{S}'(\mathbb{R}^d)$ and $\eta \in \mathcal{S}(\mathbb{R}^d)$ then $u * \eta$ can be identified with a C^∞ -function given by

$$(u * \eta)(x) = \langle u, \tau_x \tilde{\eta} \rangle,$$

namely, $\langle u * \eta, \varphi \rangle = \int_{\mathbb{R}^d} \langle u, \tau_x \tilde{\eta} \rangle \varphi(x) dx$
for $\varphi \in \mathcal{S}(\mathbb{R}^d)$.

Moreover, for every multi-index α there exist constants $C_\alpha, C'_\alpha > 0$ such that $|\partial^\alpha (u * \eta)(x)| \leq C_\alpha \cdot (1 + |x|)^{C'_\alpha}$ for all $x \in \mathbb{R}^d$.

Proof. See pp. 127-129 in [2]. ■

Definition 10. The Hilbert transform H is defined on $\mathcal{S}(\mathbb{R})$ by

$$H(\varphi) := \frac{1}{\pi} \text{p.v.} \left(\frac{1}{x} \right) * \varphi, \quad \varphi \in \mathcal{S}(\mathbb{R}).$$

3. Fourier transform.

Recall that if $f \in L^1(\mathbb{R}^d)$ then its Fourier transform is given by

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot \xi} dx, \quad \xi \in \mathbb{R}^d$$

Here, for $x = (x_1, \dots, x_d)$ and $\xi = (\xi_1, \dots, \xi_d)$, we have

$$x \cdot \xi = \sum_{j=1}^d x_j \xi_j$$

If $f \in \mathcal{S}(\mathbb{R}^d)$ then $\hat{f} \in \mathcal{S}(\mathbb{R}^d)$ and moreover,

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{i2\pi x \cdot \xi} d\xi, \quad x \in \mathbb{R}^d.$$

If $u \in \mathcal{S}'(\mathbb{R}^d)$ define $\hat{u}: \mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle \hat{u}, \varphi \rangle := \langle u, \hat{\varphi} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^d).$$

Exercise. Show that $\hat{u} \in \mathcal{S}'(\mathbb{R}^d)$.

Examples. 1) We have

$$\begin{aligned} \langle \hat{\delta}_0, \varphi \rangle &= \langle \delta_0, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int_{\mathbb{R}^d} \varphi(x) dx \\ &= \langle u_1, \varphi \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^d), \end{aligned}$$

where u_1 denotes the tempered distribution that agrees with the constant function $g(x) = 1, x \in \mathbb{R}^d$. Hence, we may write ' $\hat{\delta}_0 = 1$ '

2) If H denotes the Hilbert transform then one can show that

$$\langle \hat{H}, \varphi \rangle = \langle -i \operatorname{sgn}, \hat{\varphi} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}),$$

where sgn denotes the function given by

$$\operatorname{sgn}(\xi) := \begin{cases} 1, & \text{if } \xi > 0 \\ 0, & \text{if } \xi = 0 \\ -1, & \text{if } \xi < 0 \end{cases}.$$

See p. 316 in [2].

4) Inverse Fourier Transform.

If $u \in \mathcal{S}'(\mathbb{R}^d)$, define $\check{u}: \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathbb{C}$ by

$$\langle \check{u}, \varphi \rangle := \langle u, \check{\varphi} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^d),$$

where $\check{\varphi}(x) := \hat{\varphi}(-x), x \in \mathbb{R}^d$.

Exercise. Show that $\check{u} \in \mathcal{S}'(\mathbb{R}^d)$

Proposition 10. Let $k \in L^1(\mathbb{R}^d)$ be such that

$$\int_{\mathbb{R}^d} k(x) dx = 1.$$

Consider the approximate identity $(k_t)_{t>0}$ where $k_t(x) := t^{-d} k(t^{-1}x)$, $x \in \mathbb{R}^d$, $t > 0$.

Then, $k_t \xrightarrow[t \rightarrow 0^+]{} \delta_0$ in $\mathcal{S}'(\mathbb{R}^d)$ that is,

$$\lim_{t \rightarrow 0^+} \langle k_t, \varphi \rangle = \varphi(0) = \langle \delta_0, \varphi \rangle$$

for all $\varphi \in \mathcal{S}(\mathbb{R}^d)$.

Proof. Fix a $\varphi \in \mathcal{S}(\mathbb{R}^d)$ and note that for $t > 0$ one has

$$\begin{aligned} \langle k_t, \varphi \rangle &= \int_{\mathbb{R}^d} k_t(x) \varphi(x) dx = \int_{\mathbb{R}^d} t^{-d} k(t^{-1}x) \varphi(x) dx \\ &= \int_{\mathbb{R}^d} k(y) \varphi(ty) dy. \end{aligned}$$

By the dominated convergence theorem,

$$\begin{aligned} \lim_{t \rightarrow 0^+} \langle k_t, \varphi \rangle &= \int_{\mathbb{R}^d} \lim_{t \rightarrow 0^+} k(y) \varphi(ty) dy = \int_{\mathbb{R}^d} k(y) \varphi(0) dy \\ &= \varphi(0) \int_{\mathbb{R}^d} k(y) dy \\ &= \varphi(0) = \delta_0(\varphi). \end{aligned}$$

The notes of this lecture are based on the following references:

- [1] Duoandikostzea, Javier. Fourier Analysis. Translated and revised from the 1995 Spanish original by David Cruz-Uribe. Graduate Studies in Mathematics, 29. American Mathematical Society, Providence, RI, 2001, xviii + 222 pp.
- [2] Grafakos, Loukas. Classical Fourier Analysis, Third Edition. Graduate Texts in Mathematics, 249. Springer, New York, 2014. xviii + 638 pp.